

Hurdle Models

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Introduction

Hurdle models split count outcomes into two distinct processes: a Bernoulli decision for zero versus non-zero, and a zero-truncated count distribution for the positive counts. They are appropriate when the question of “did anything happen?” is conceptually different from “given that something happened, how much?” — for example, healthcare visits (decided to see a doctor at all, then number of visits), insurance claims, hospital admissions for a rare condition.

Prerequisites

A working understanding of Poisson and negative binomial regression, the zero-inflated alternative, and the conceptual distinction between two-part and mixture models for zero-heavy counts.

Theory

The hurdle model is

$$P(Y = 0) = 1 - \pi, \quad P(Y = y) = \pi \cdot \frac{f(y)}{1 - f(0)}, \quad y \geq 1,$$

with π the probability of clearing the hurdle (logistic regression on covariates) and $f(y)$ a count distribution (Poisson or NB) truncated at zero. The two parts are estimated jointly by maximum likelihood; covariates can differ between the zero and count submodels.

The contrast with zero-inflated models is conceptual: hurdle treats *all* zeros as arising from one Bernoulli process, while zero-inflated treats some zeros as “structural” (always zero) and others as “sampling” (could have been positive but happened to be zero). Choose based on the scientific story.

Assumptions

A count outcome with a meaningful two-stage data-generating process; the chosen count distribution (Poisson or NB) is appropriate for the positive counts.

R Implementation

```
library(pscl)

set.seed(2026)
d <- data.frame(x = rnorm(300))
d$y <- ifelse(rbinom(300, 1, plogis(0 + 0.4 * d$x)) == 0, 0,
             rpois(300, lambda = exp(1 + 0.5 * d$x)))

fit_hurdle <- hurdle(y ~ x | x, data = d)
summary(fit_hurdle)
```

```
# Zero-inflated for comparison
fit_zi <- zeroinfl(y ~ x | x, data = d)
AIC(fit_hurdle, fit_zi)
```

Output & Results

`summary(fit_hurdle)` returns two coefficient tables: the count part (zero-truncated Poisson or NB) and the zero hurdle (binomial). Each part has its own intercept and slope estimates with Wald z -tests. The AIC comparison against `zeroinfl` quantifies which framework better matches the data.

Interpretation

A reporting sentence: “A hurdle model decomposed the data into a binomial hurdle (each unit increase in x raised the log-odds of any positive count by 0.40, $p < 0.001$) and a zero-truncated Poisson count model (rate ratio 1.65 per unit increase in x among positive counts, 95 % CI 1.40 to 1.95). Hurdle was favoured over zero-inflated by AIC ($\Delta = 4$).” Always describe both parts and use the AIC comparison to justify the framework choice.

Practical Tips

- Hurdle is cleaner than zero-inflated when the scientific story has a clear two-stage interpretation; mixing the two approaches without justification confuses readers.
- For overdispersed positive counts, use `dist = "negbin"` in the count part; this is essential for most real count data.
- Different covariates can drive the two parts; use `y ~ x_count | x_zero` to specify them separately.
- Zero-inflated assumes some zeros are structural and could not have been any other value; hurdle assumes all zeros come from one decision process.
- AIC and BIC compare hurdle and zero-inflated objectively when both are well-fitted; do not rely on visual inspection alone.
- For hierarchical hurdle models, `glmmTMB(family = truncated_nbinom2(), ziformula = ~ ...)` provides random-effect extensions.

R Packages Used

`pscl::hurdle()` for canonical hurdle models with Poisson, NB, or geometric count parts; `glmmTMB` for hurdle and zero-inflated mixed-effects extensions; `countreg` for additional count-model variants and rootogram diagnostics; `DHARMA` for residual diagnostics across hurdle and zero-inflated families.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors

- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI